

Statistical Physics

Course notes

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1 A Macroscopic View of Matter

1.2 Thermodynamics

A macroscopic system left to itself relaxes to a stationary state, called **thermal equilibrium**.

- **Thermodynamics** is a phenomenological theory of equilibrium states and of the transitions among them.
- **Statistical mechanics** is concerned with deducing the thermodynamic properties of a macroscopic system from its microscopic structure.
- **Kinetic theory** aims at a microscopic description of the transition process between equilibrium states.

Extensive quantities are proportional to the amount of substance present. **Intensive quantities** are independent of the amount of substance present.

1.3 The Thermodynamic Limit

Consider N particles in a volume V . The **thermodynamic limit** is

$$N \rightarrow \infty, \quad V \rightarrow \infty, \quad \frac{N}{V} = \text{fixed},$$

in which the system becomes translationally invariant.

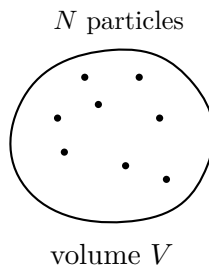


Figure 1.1: N particles in a volume V .

The **thermodynamic state** is specified by a number of thermodynamic variables, which are assumed to be either extensive (proportional to N) or intensive (independent of N). In the thermodynamic limit one must, mathematically speaking, only use intensive quantities.

1.4 Thermodynamic Transformations

When a body is in thermal equilibrium, the thermodynamic variables are not independent of one another; they satisfy an **equation of state**

$$f(P, V, T) = 0, \tag{1.1}$$

where f is continuously differentiable, except possibly at special points.

A change in the external conditions changes the equilibrium state of the system: this is a **thermodynamic transformation**.

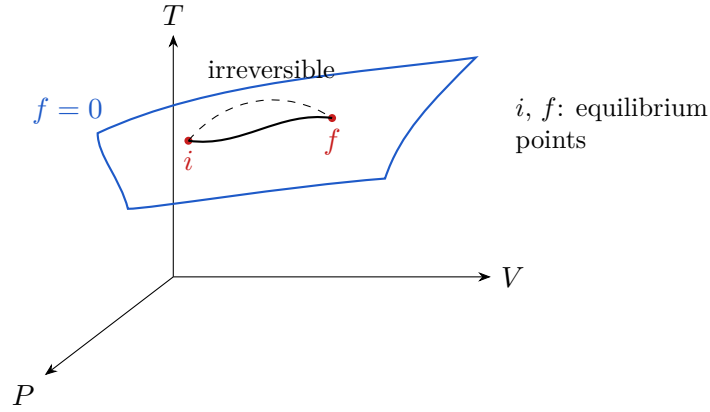


Figure 1.2: Transformation between two equilibrium states i and f on the surface $f(P, V, T) = 0$. A quasi-static path lies on the surface; an irreversible one (dashed) does not.

If the transformation proceeds *slowly*, the system can be considered to remain at equilibrium at every step: this is a **quasi-static** transformation, which usually means that the transformation is **reversible**. In a reversible transformation the mechanical work done by the system is

$$dW = P dV.$$

Between two states A and B ,

$$\Delta W = \int_A^B P dV,$$

which is path dependent: it is the area underneath the path followed by the system in the P - V plane. If the path is closed (a cycle), the work done is the area enclosed.

Heat transfer is a uniquely thermodynamic process. Heat is a form of energy, and phenomenologically

$$\Delta Q := C \Delta T,$$

where C is the **heat capacity**. The intensive heat capacities (C/n , C/N , C/V) are called **specific heats**.

The heat absorbed by the body, like the work done by it, depends on the path of the transformation. Some common processes are:

- isothermal (T constant);
- isobaric (P constant);
- isochoric (V constant);
- adiabatic ($\Delta Q = 0$).

1.5 Classical Ideal Gas

The **ideal gas** is the limit of low density and high temperature of a gas. Its equation of state is

$$PV = Nk_B T, \quad Nk_B = nR, \quad (1.2)$$

where n is the number of moles, R the gas constant and k_B is Boltzmann's constant,

$$k_B = 1.380\,649 \times 10^{-23} \text{ J/K}.$$

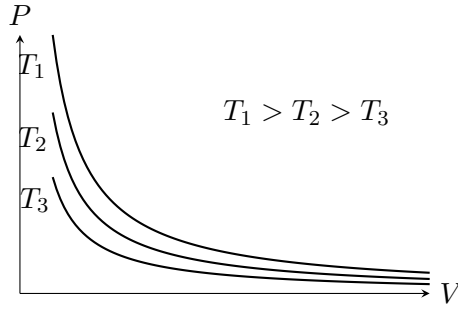


Figure 1.3: Isotherms of the ideal gas.

1.6 First Law of Thermodynamics

$$dU = dQ - dW, \quad (1.3)$$

where U is the **internal energy**, dQ is the heat absorbed by the system and dW the work done by the system. dU is an **exact differential**, whereas dQ and dW each depend on the path.

2 Heat and Entropy

2.1 The Heat Equations

Recall that, by the equation of state (1.1), only two of the variables P, V, T are independent. Hence

$$\begin{aligned} dU(P, V) &= \left(\frac{\partial U}{\partial P}\right)_V dP + \left(\frac{\partial U}{\partial V}\right)_P dV, \\ dU(P, T) &= \left(\frac{\partial U}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial T}\right)_P dT, \\ dU(V, T) &= \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT. \end{aligned}$$

Using the first law in the form $dQ = dU + P dV$,

$$\begin{aligned} dQ(P, V) &= \left(\frac{\partial U}{\partial P}\right)_V dP + \left[\left(\frac{\partial U}{\partial V}\right)_P + P\right] dV, \\ dQ(P, T) &= \left(\frac{\partial U}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial T}\right)_P dT + P dV, \\ dQ(V, T) &= \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right] dV + \left(\frac{\partial U}{\partial T}\right)_V dT. \end{aligned}$$

Writing $dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT$ in the second line,

$$dQ(P, T) = \left[\left(\frac{\partial U}{\partial P}\right)_T + P \left(\frac{\partial V}{\partial P}\right)_T\right] dP + \left(\frac{\partial}{\partial T}(U + PV)\right)_P dT.$$

The combination in the last term defines the **enthalpy**

$$H := U + PV, \quad C_P = \left(\frac{\partial H}{\partial T}\right)_P.$$

2.2 Applications to the Ideal Gas

Free expansion. A gas at temperature T initially occupies part of a thermally insulated box, the rest being vacuum; the partition is removed and the gas fills the box.

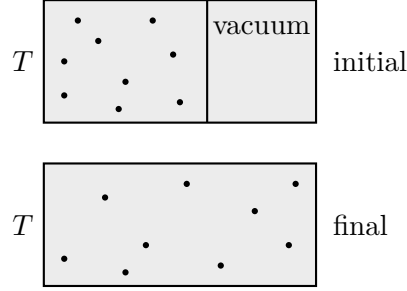


Figure 2.1: Free expansion of an ideal gas.

Since $dQ = 0$ and $dW = 0$, we have $dU = 0$, and experimentally the temperature does not change:

$$U(V_1, T) = U(V_2, T) \implies U = U(T).$$

Therefore

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{dU}{dT}, \quad U(T) = C_V T,$$

and, using (1.2),

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P = \left(\frac{\partial}{\partial T} (U + PV) \right)_P = C_V + Nk_B,$$

that is,

$$C_P - C_V = Nk_B.$$

Adiabatic process. For an adiabatic process $dQ = 0$, so $dU = -P dV$, i.e.

$$C_V dT + P dV = 0.$$

With $PV = Nk_B T$,

$$\begin{aligned} \frac{C_V}{Nk_B} d(PV) + P dV &= 0, \\ C_V(P dV + V dP) + Nk_B P dV &= 0, \\ (C_V + Nk_B) P dV + C_V V dP &= 0, \\ C_P P dV + C_V V dP &= 0, \end{aligned}$$

so that

$$\frac{dP}{P} + \gamma \frac{dV}{V} = 0, \quad \gamma := \frac{C_P}{C_V} > 1,$$

and therefore

$$PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}.$$

Since $\gamma > 1$, adiabats are steeper than isotherms in the P - V plane.

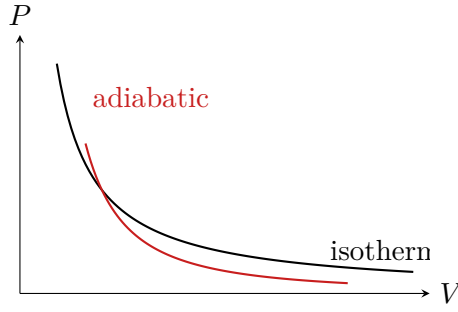


Figure 2.2: An adiabat ($PV^\gamma = \text{const}$) is steeper than an isotherm.

2.3 Carnot Cycle

In a cyclic transformation the final state equals the initial one, so $\Delta U = 0$ and

$$\Delta Q = \Delta W = \oint P dV = \text{area enclosed}.$$

The heat absorbed over the cycle is transformed into work.

A **Carnot cycle** consists of two isotherms at $T_2 > T_1$ and two adiabats. Along the upper isotherm the system absorbs heat Q_2 from the hot reservoir, and along the lower one it rejects heat Q_1 to the cold reservoir.

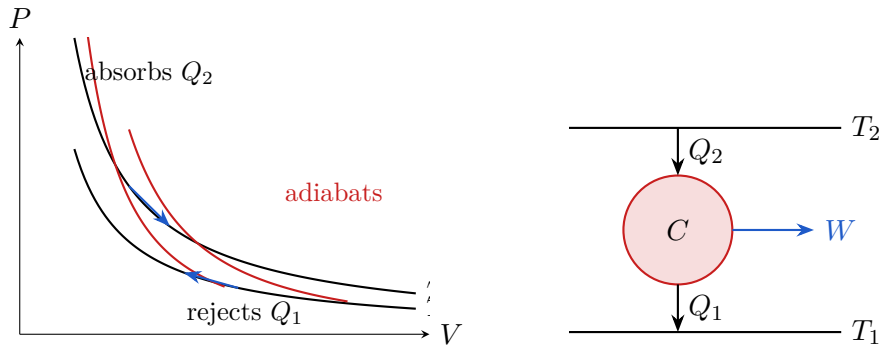


Figure 2.3: Carnot cycle in the P - V plane (left) and schematic Carnot engine operating between reservoirs at $T_2 > T_1$ (right).

The work done is $W = Q_2 - Q_1$, and the **efficiency of the Carnot engine** is

$$\eta := \frac{W}{Q_2} = 1 - \frac{Q_1}{Q_2}. \quad (2.1)$$

2.4 Second Law of Thermodynamics

Statement (Clausius). There does not exist a thermodynamic transformation whose *sole* effect is to deliver heat from a reservoir of lower temperature to a reservoir of higher temperature.

Statement (Kelvin). There does not exist a thermodynamic transformation whose *sole* effect is to extract heat from a reservoir and convert it entirely into work.

The two statements are equivalent.

2.5 Absolute Temperature

By the second law, a Carnot engine cannot be 100% efficient. We define the **absolute temperature** θ of a heat reservoir such that the ratio of the absolute temperatures of two reservoirs is given by

$$\frac{\theta_1}{\theta_2} := \frac{Q_1}{Q_2} = 1 - \eta,$$

with $Q_1 > 0$ by the second law. Absolute zero, $\theta = 0$, is a limiting value which, according to the second law, can never be reached.

The absolute temperature coincides with the ideal gas temperature, $\theta = T$, as can be proven by running a Carnot cycle with an ideal gas.

2.6 Temperature as Integrating Factor

In an arbitrary cyclic process $\mathcal{P}$ the following inequality holds:

$$\oint_{\mathcal{P}} \frac{\mathrm{d}Q}{T} \leq 0, \quad (2.2)$$

where the equality holds if $\mathcal{P}$ is reversible.

Proof. Divide the cycle into k steps. At step i the system is at temperature T_i and absorbs heat Q_i , supplied by a Carnot engine C_i operating between a common reservoir at T_0 and T_i ; C_i extracts $Q_i^{(0)}$ from the reservoir.

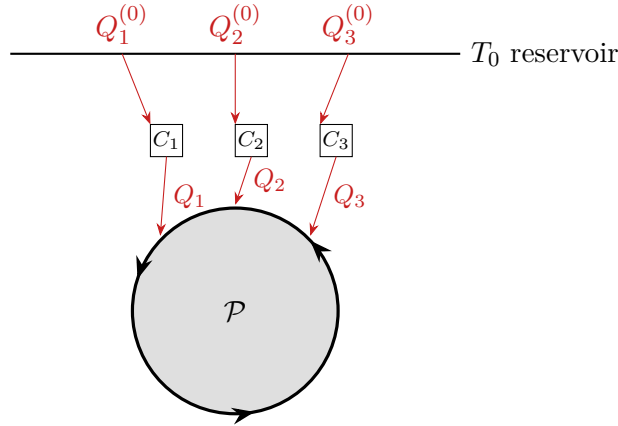


Figure 2.4: The cycle $\mathcal{P}$ exchanges heat Q_i at temperatures T_i through Carnot engines C_i fed by a single reservoir at T_0 .

By (2.1) and the definition of absolute temperature,

$$W = \sum_{i=1}^k Q_i, \quad \frac{Q_i^{(0)}}{Q_i} = \frac{T_0}{T_i}.$$

The total heat extracted from the reservoir and the total work done are

$$Q_{\text{tot}} = \sum_{i=1}^k Q_i^{(0)} = T_0 \sum_{i=1}^k \frac{Q_i}{T_i},$$

$$W_{\text{tot}} = W + \sum_{i=1}^k (Q_i^{(0)} - Q_i) = \sum_{i=1}^k Q_i^{(0)} = T_0 \sum_{i=1}^k \frac{Q_i}{T_i}.$$

The net effect of the whole process is to extract heat from a single reservoir and convert it into work, which by the (Kelvin) second law requires $W_{\text{tot}} \leq 0$. Taking $k \rightarrow \infty$,

$$\lim_{k \rightarrow \infty} \sum_{i=1}^k \frac{Q_i}{T_i} = \oint_{\mathcal{P}} \frac{dQ}{T} \leq 0.$$

If $\mathcal{P}$ is reversible, running it backwards gives the opposite inequality, so $\oint_{\mathcal{P}} dQ/T = 0$. Hence, for reversible processes, dQ/T is an exact differential.

2.7 Entropy

The **entropy** S is defined through the exact differential

$$dS := \frac{dQ}{T},$$

so that S is a **state function**. For a reversible path from A to B ,

$$\Delta S = S(B) - S(A) := \int_A^B \frac{dQ}{T}.$$

Now let $\mathcal{P}$ be an arbitrary path from A to B and R a reversible one.

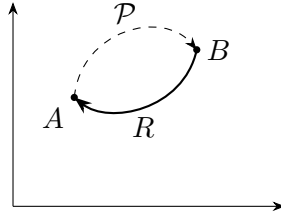


Figure 2.5: An arbitrary path $\mathcal{P}$ from A to B , closed by a reversible path R back to A .

Applying (2.2) to the cycle $\mathcal{P} + R$,

$$\oint_{\mathcal{P}+R} \frac{dQ}{T} \leq 0 \quad \Rightarrow \quad \int_{\mathcal{P}} \frac{dQ}{T} - \underbrace{\int_R \frac{dQ}{T}}_{\Delta S} \leq 0,$$

where R is traversed from A to B in the second integral. Therefore

$$\Delta S \geq \int_{\mathcal{P}} \frac{dQ}{T}, \quad \begin{cases} \Delta S = \int_{\mathcal{P}} dQ/T & \text{if } \mathcal{P} \text{ is reversible,} \\ \Delta S > \int_{\mathcal{P}} dQ/T & \text{if } \mathcal{P} \text{ is irreversible,} \\ \Delta S \geq 0 & \text{if } \mathcal{P} \text{ is adiabatic.} \end{cases}$$

In particular, for an isolated system (every process is adiabatic)

$$\Delta S \geq 0.$$

5 The Statistical Approach

5.2 Random Walk

“Thermodynamic order emerges precisely from randomness.”

Let $W(k, n)$ be the probability that, after taking n random steps of unit length (forwards or backwards with equal probability), the particle is k steps from the starting point, $-n \leq k \leq n$. There are 2^n possible trajectories. If F steps are taken forwards and B backwards,

$$F + B = n, \quad F - B = k \quad \implies \quad F = \frac{n+k}{2}, \quad B = \frac{n-k}{2}.$$

The number of ways of choosing the F forward steps among the n steps is $\binom{n}{F} = \frac{n!}{F!(n-F)!}$, hence

$$W(k, n) = \frac{1}{2^n} \binom{n}{F} = \frac{2^{-n} n!}{\left[\frac{n+k}{2}\right]! \left[\frac{n-k}{2}\right]!}.$$

For $n \gg 1$ and $n \gg |k|$, using Stirling's formula

$$\ln(n!) \simeq n \ln n - n + \ln \sqrt{2\pi n},$$

one finds

$$W(k, n) \simeq \frac{1}{\sqrt{2\pi n}} e^{-k^2/2n}.$$

Introducing the step length x_0 and the time per step t_0 , with $x = kx_0$ and $t = nt_0$, the continuum limit gives the diffusion kernel

$$W(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-x^2/4Dt}, \quad D = \frac{x_0^2}{2t_0}, \quad \int_{-\infty}^{\infty} dx W(x, t) = 1.$$

5.4 Distribution Function

The state of an N -particle system is a point in the $6N$ -dimensional **Γ -space**; as the system evolves at fixed energy, this point moves on the **energy surface**. Alternatively, the state of each single particle is a point in the 6-dimensional **μ -space**, and the system is a cloud of N points in it.

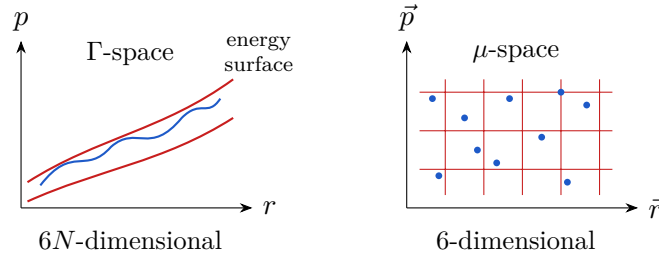


Figure 5.1: Γ -space (one point for the whole system) and μ -space (one point per particle), the latter divided into cells.

The μ -space is divided into 6-dimensional volume elements

$$\Delta\tau = \Delta p_x \Delta p_y \Delta p_z \Delta x \Delta y \Delta z,$$

each one an “infinitesimal” cell that nevertheless contains a large number of atoms. The number of atoms in cell i is the **occupation number** n_i . The **distribution function** $f(\vec{p}, \vec{r}, t)$ is the occupation number per unit volume of μ -space:

$$f(\vec{p}_i, \vec{r}_i, t) \Delta\tau = n_i,$$

subject to the restrictions

$$\sum_i n_i = N, \quad \sum_i n_i \varepsilon_i = E.$$

Taking $\Delta\tau \rightarrow d^3p d^3r$,

$$\int d^3p d^3r f(\vec{p}, \vec{r}, t) = N, \quad \int d^3p d^3r f(\vec{p}, \vec{r}, t) \varepsilon = E,$$

with $\varepsilon = p^2/2m$ for an ideal gas. If the density is uniform,

$$\int d^3p f(\vec{p}, t) = \frac{N}{V}, \quad \int d^3p f(\vec{p}, t) \varepsilon = \frac{E}{V}.$$

5.5 Ergodic Hypothesis

Statement. Given sufficiently long time, the representative point of an isolated system will come arbitrarily close to any given point on the energy surface.

5.6 Statistical Ensemble

A **statistical ensemble** is, conceptually, an infinite collection of identical copies of the system, characterised by a density function $\rho(p, q, t)$ in Γ -space:

$$\rho(p, q, t) dp dq = \text{number of systems in } dp dq \text{ at time } t,$$

where $dp = d^{3N}p$ and $dq = d^{3N}q$ ($6N$ -dimensional). The **probability density** is

$$\frac{\rho(p, q, t)}{\int dp dq \rho(p, q, t)},$$

and the **ensemble average** of a physical quantity $O(p, q)$ is defined as

$$\langle O \rangle := \frac{\int dp dq O(p, q) \rho(p, q, t)}{\int dp dq \rho(p, q, t)},$$

to be evaluated in the thermodynamic limit. In thermal equilibrium ρ does not depend explicitly on t , and $\rho = \rho(H(p, q))$.

5.7 Microcanonical Ensemble

$$\rho(H(p, q)) = \begin{cases} 1, & E < H(p, q) < E + \Delta, \\ 0, & \text{otherwise,} \end{cases}$$

where $\Delta \ll E$ is the tolerance of energy measurements. The volume of Γ -space occupied by the ensemble is

$$\Gamma(E, V) = \int dp dq \rho(H(p, q)) = \int_{E < H(p, q) < E + \Delta} dp dq.$$

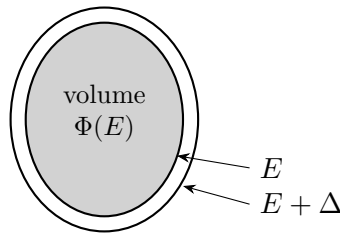


Figure 5.2: The energy shell $E < H < E + \Delta$ in Γ -space.

With $\Phi(E, V)$ the volume enclosed by the energy surface,

$$\Phi(E, V) = \int_{H(p, q) < E} dp dq, \quad \Gamma(E, V) \simeq \frac{\partial \Phi}{\partial E} \Delta.$$

The **connection to thermodynamics** is made through the definition of entropy

$$S(E, V) := k_B \ln \Gamma(E, V),$$

with $k_B = 1.380\,649 \times 10^{-23}$ J/K the Boltzmann constant. The logarithm makes S additive for independent systems.

5.8 Correct Boltzmann Counting

In defining the distribution function we have discretised the μ -space of a single atom into cells labelled by i , and denoted the occupation number of cell i by n_i . The set $\{n_i\}$ is a **distribution**. In classical physics, a permutation of the coordinates of two atoms leads to a new microscopic state. Let

$$\Omega\{n_i\} := \Omega\{n_1, n_2, \dots\}$$

be the volume in Γ -space corresponding to the distribution $\{n_i\}$. Then

$$\Gamma(E, V) = \sum_{\{n_i\}} \Omega\{n_1, n_2, \dots\},$$

where the sum runs over all sets satisfying $\sum_i n_i = N$ and $\sum_i n_i \varepsilon_i = E$.

1	2	3		...	K
n_1	n_2	n_3		...	n_K

Figure 5.3: Cells of μ -space and their occupation numbers.

The ensemble average of an occupation number is

$$\langle n_k \rangle = \frac{\sum_{\{n_i\}} n_k \Omega\{n_i\}}{\sum_{\{n_i\}} \Omega\{n_i\}}.$$

Counting all permutations of distinguishable atoms gives

$$\Omega_0\{n_i\} = \frac{N!}{n_1! n_2! \dots n_K!} \lambda_1^{n_1} \lambda_2^{n_2} \dots \lambda_K^{n_K},$$

where λ_i is the intrinsic probability of finding a particle in cell i . The factor $N!$ must be dropped: this accounts for the **indistinguishability of particles**, which is a quantum effect with no classical analogue. The correct Boltzmann counting is therefore

$$\Omega\{n_i\} = \frac{1}{n_1! \dots n_K!} \lambda_1^{n_1} \dots \lambda_K^{n_K}.$$

5.9 Distribution Entropy: Boltzmann's H

The **distribution entropy** is

$$H_B\{n_i\} := k_B \ln \Omega\{n_i\},$$

with

$$\ln \Omega\{n_i\} = - \sum_i \ln(n_i!) + \sum_i n_i \ln \lambda_i.$$

Using Stirling's approximation $\ln(n!) \simeq n \ln n - n$ (and dropping the terms linear in n_i , which are fixed by the constraints),

$$H_B\{n_i\} \simeq -k_B \sum_i n_i \ln n_i.$$

On the other hand,

$$\frac{S}{k_B} = \ln \left(\sum_{\{n_i\}} \Omega\{n_i\} \right) = \ln \left(\sum_{\{n_i\}} e^{H_B\{n_i\}/k_B} \right).$$

In the thermodynamic limit the sum over distributions is dominated by the most probable distribution $\{\bar{n}_i\}$, so

$$S \simeq H_B\{\bar{n}_i\} = -k_B \sum_i \bar{n}_i \ln \bar{n}_i.$$

5.10 The Most Probable Distribution

The most probable distribution $\{\bar{n}_i\}$ coincides with the ensemble average $\langle n_i \rangle$ in the thermodynamic limit. It is found from

$$\delta H_B|_{\{\bar{n}_i\}} = 0, \quad \text{with} \quad \sum_i \delta n_i = 0, \quad \sum_i \varepsilon_i \delta n_i = 0.$$

Using Lagrange multipliers α and β ,

$$\delta \left[\frac{H_B\{n_i\}}{k_B} + \alpha \sum_i n_i - \beta \sum_i \varepsilon_i n_i \right] = 0,$$

that is (keeping the λ_i),

$$\sum_i [-\ln n_i + \alpha - \beta \varepsilon_i + \ln \lambda_i] \delta n_i = 0 \implies \ln \bar{n}_i = \alpha - \beta \varepsilon_i + \ln \lambda_i.$$

Taking all intrinsic probabilities equal,

$$\bar{n}_i = C e^{-\beta \varepsilon_i}, \quad \sum_i C e^{-\beta \varepsilon_i} = N, \quad \sum_i C e^{-\beta \varepsilon_i} \varepsilon_i = E.$$

In the limit $N \rightarrow \infty$, $n_i \rightarrow f(\vec{p}_i, \vec{r}_i) d^3p d^3r$. For a **free gas** in the absence of external potentials this gives the **Maxwell–Boltzmann distribution**

$$f(\vec{p}) = C e^{-\beta p^2/2m}, \tag{5.1}$$

with

$$\int d^3p f(\vec{p}) = \frac{N}{V} = n, \quad \frac{1}{n} \int d^3p f(\vec{p}) \frac{p^2}{2m} = \frac{E}{N}.$$

6 Maxwell–Boltzmann Distribution

6.1 Determining the Parameters

Starting from (5.1), $f(\vec{p}) = C e^{-\beta p^2/2m}$, the two conditions

$$\int d^3p f(\vec{p}) = \frac{N}{V}, \quad \frac{1}{n} \int d^3p f(\vec{p}) \frac{p^2}{2m} = \frac{E}{N}$$

give

$$C = n \left(\frac{\beta}{2\pi m} \right)^{3/2}, \quad \beta = \frac{3}{2} \frac{N}{E}.$$

We will see below that $\beta = 1/k_B T$.

6.2 Pressure of an Ideal Gas

The pressure on a wall perpendicular to x is

$$\text{pressure} = (\text{momentum transfer per collision}) \times (\text{flux of atoms}) = (2mv_x) \times (v_x f(\vec{p}) d^3p),$$

integrated over atoms moving towards the wall:

$$P = \int_{v_x > 0} d^3p f(\vec{p}) v_x (2mv_x).$$

By isotropy $v_x^2 \rightarrow \frac{1}{3}(v_x^2 + v_y^2 + v_z^2) = p^2/3m^2$, so

$$P = \frac{2}{3} \frac{E}{V}.$$

Comparing with the ideal gas law $PV = Nk_B T$,

$$\frac{N}{V} k_B T = \frac{2}{3} \frac{E}{V} \implies \frac{2}{3} \frac{E}{N} = k_B T,$$

and therefore

$$\beta = \frac{1}{k_B T}.$$

The parameters of the Maxwell–Boltzmann distribution are thus

$$\beta = \frac{1}{k_B T}, \quad C = \frac{n}{(2\pi m k_B T)^{3/2}}.$$

6.3 Equipartition of Energy

Each quadratic term in the energy (each **degree of freedom**) contributes $\frac{1}{2}k_B T$ to the internal energy per particle.

6.4 Entropy

From $S = k_B \ln \Gamma$ and Section 5.9, S is given by the most probable distribution,

$$S(V, T) = -k_B \sum_i \bar{n}_i \ln \bar{n}_i, \quad \bar{n}_i = f_i \Delta\tau,$$

so that

$$S(V, T) = -V k_B \int d^3p f(\vec{p}) \ln f(\vec{p}).$$

Evaluating the integral with (5.1),

$$S(V, T) = N k_B \ln \left(\frac{T^{3/2}}{n} \right) + S_0,$$

which is extensive, as it should be.

6.6 Derivation of Thermodynamics

For the ideal gas we have obtained

$$PV = Nk_B T, \quad U(T) = E = \frac{3}{2} Nk_B T,$$

so that

$$C_V = \frac{3}{2} Nk_B, \quad PV = \frac{2}{3} U, \quad \frac{S}{k_B} = N \ln V + \frac{3}{2} N \ln T + \text{const.}$$

Differentiating, and using $dT/T = dU/U$ together with $dV/V = P dV/(Nk_B T)$,

$$\frac{dS}{Nk_B} = \frac{dV}{V} + \frac{3}{2} \frac{dU}{U} = \frac{P dV}{Nk_B T} + \frac{3}{2} \frac{dU}{Nk_B T},$$

i.e.

$$T dS = dU + P dV,$$

which is the first law with $dQ = T dS$.

6.7 Fluctuations

How probable is the most probable distribution? With

$$\Omega\{n_i\} = \frac{1}{n_1! n_2! \dots n_K!} \lambda_1^{n_1} \lambda_2^{n_2} \dots \lambda_K^{n_K},$$

we have

$$\langle n_k \rangle = \frac{\sum_{\{n_i\}} n_k \Omega\{n_i\}}{\sum_{\{n_i\}} \Omega\{n_i\}} = \lambda_k \frac{\partial}{\partial \lambda_k} \ln \left(\sum_{\{n_i\}} \Omega\{n_i\} \right).$$

Writing $\Gamma = \sum_{\{n_i\}} \Omega\{n_i\}$ and $\nu_i := \ln \lambda_i$,

$$\begin{aligned} \langle n_k \rangle &= \frac{\partial}{\partial \nu_k} \ln \Gamma, \\ \langle n_k^2 \rangle &= \frac{1}{\Gamma} \frac{\partial^2 \Gamma}{\partial \nu_k^2} = \frac{\partial}{\partial \nu_k} \left(\frac{1}{\Gamma} \frac{\partial \Gamma}{\partial \nu_k} \right) + \frac{1}{\Gamma^2} \left(\frac{\partial \Gamma}{\partial \nu_k} \right)^2 = \frac{\partial \langle n_k \rangle}{\partial \nu_k} + \langle n_k \rangle^2, \end{aligned}$$

so that

$$\langle n_k^2 \rangle - \langle n_k \rangle^2 = \frac{\partial \langle n_k \rangle}{\partial \nu_k} = \lambda_k \frac{\partial}{\partial \lambda_k} \langle n_k \rangle.$$

Assuming $\langle n_k \rangle \simeq \bar{n}_k = \lambda_k C e^{-\beta \varepsilon_k}$,

$$\langle n_k^2 \rangle - \langle n_k \rangle^2 = \langle n_k \rangle \implies \frac{\langle n_k^2 \rangle - \langle n_k \rangle^2}{N^2} = \left\langle \frac{n_k}{N} \right\rangle \frac{1}{N}.$$

As $N \rightarrow \infty$ the relative fluctuations vanish and $\langle n_k \rangle \simeq \bar{n}_k$: the Maxwell–Boltzmann distribution is the overwhelmingly prevalent condition for a gas in equilibrium.

6.8 The Boltzmann Factor

If a system has possible states labelled by i , and the energy of state i is ε_i , then the probability of finding the system in state i is proportional to the **Boltzmann factor** $e^{-\varepsilon_i/k_B T}$. The absolute probability of the occurrence of state i is

$$\frac{e^{-\varepsilon_i/k_B T}}{\sum_j e^{-\varepsilon_j/k_B T}}.$$

8 Canonical Ensemble

8.1 Review of the Microcanonical Ensemble

All members of the microcanonical ensemble have the same energy E , within a small tolerance. The entropy is

$$S(E) = k_B \ln \Gamma(E),$$

where $\Gamma(E)$ is the total number of states of the system with energy E , and the temperature follows from

$$\frac{1}{T} = \frac{\partial S}{\partial E}.$$

8.2 Classical Canonical Ensemble

Now the energy is not fixed: it is exchanged with the environment, a **reservoir**. There is no exchange of particles, only of energy.

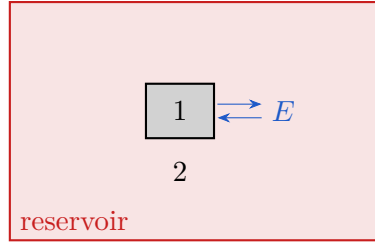


Figure 8.1: System 1 exchanges energy with the reservoir 2.

The total system is isolated, with Hamiltonian

$$H(p_1, q_1, p_2, q_2) = H_1(p_1, q_1) + H_2(p_2, q_2),$$

and

$$N = N_1 + N_2, \quad N_1 \ll N_2, \quad E = E_1 + E_2, \quad E_1 \ll E_2.$$

The phase-space volume of the reservoir is

$$\Gamma_2(E_2) = \int_{E_2 < H_2 < E_2 + \Delta} dp_2 dq_2.$$

The probability that system 1 is in $dp_1 dq_1$ is proportional to $dp_1 dq_1 \Gamma_2(E - E_1)$, so

$$\rho_1(p_1, q_1) = \Gamma_2(E - E_1) = \Gamma_2(E - H_1(p_1, q_1)).$$

Expanding the entropy of the reservoir,

$$k_B \ln \Gamma_2(E - E_1) = S_2(E - E_1) \simeq S_2(E) - E_1 \left. \frac{\partial S_2}{\partial E} \right|_E = S_2(E) - \frac{E_1}{T},$$

hence

$$\rho_1(p_1, q_1) = e^{S_2(E)/k_B} e^{-H_1(p_1, q_1)/k_B T}.$$

Dropping the constant prefactor, the **canonical ensemble** is

$$\rho(p, q) = e^{-\beta H(p, q)}.$$

8.3 The Partition Function

$$Q_N(T, V) := \int \frac{dp dq}{\mathcal{Z}} e^{-\beta H(p, q)}, \quad \mathcal{Z} = N! h^{3N}. \quad (8.1)$$

The factor $N!$ is the correct Boltzmann counting, and h^{3N} makes the measure $dp dq / \mathcal{Z}$ dimensionless.

8.4 Connection with Thermodynamics

The internal energy is

$$U = \langle H \rangle = \frac{\int dp dq H e^{-\beta H}}{\int dp dq e^{-\beta H}} = -\frac{\partial}{\partial \beta} \ln Q_N.$$

The general connection between the canonical ensemble and thermodynamics is given by the statement

$$Q_N(T, V) = e^{-\beta A(V, T)},$$

where $A = U - TS$ is the **Helmholtz free energy**. To check it, write

$$\int \frac{dp dq}{\mathcal{Z}} e^{\beta(A-H)} = 1$$

and differentiate with respect to β :

$$\int \frac{dp dq}{\mathcal{Z}} e^{\beta(A-H)} \left[A + \beta \frac{\partial A}{\partial \beta} - H \right] = 0 \implies A - T \frac{\partial A}{\partial T} - U = 0,$$

which is consistent with

$$S = -\frac{\partial A}{\partial T} \quad \text{and} \quad A = U - TS.$$

8.6 Minimisation of the Free Energy

Inserting $1 = \int dE \delta(H(p, q) - E)$ in (8.1),

$$Q_N(T, V) = \int dE e^{-\beta E} \omega(E), \quad \omega(E) = \int \frac{dp dq}{\mathcal{Z}} \delta(H(p, q) - E),$$

with $S(E) = k_B \ln \omega(E)$. Thus

$$Q_N(T, V) = \int dE \omega(E) e^{-\beta E} = \int dE e^{-\beta(E - TS(E))} = \int dE e^{-\beta A(E)}, \quad A(E) = E - TS(E).$$

The dominant contribution comes from the minimum of $A(E)$:

$$\frac{\partial A}{\partial E} = 1 - T \frac{\partial S}{\partial E} = 0 \implies \frac{\partial S}{\partial E} = \frac{1}{T},$$

$$\frac{\partial^2 A}{\partial E^2} = -T \frac{\partial^2 S}{\partial E^2} = -T \frac{\partial}{\partial E} \left(\frac{1}{T} \right) = \frac{1}{T} \frac{\partial T}{\partial E} = \frac{1}{TC_V} > 0,$$

so it is indeed a minimum. Expanding around it, at $E = \bar{E}$,

$$A(E) = A(\bar{E}) + \frac{1}{2TC_V} (E - \bar{E})^2 + \dots,$$

$$Q_N(T, V) \simeq e^{-\beta A(\bar{E})} \int dE e^{-\frac{\beta}{2TC_V} (E - \bar{E})^2} = \sqrt{2\pi k_B T^2 C_V} e^{-\beta A(\bar{E})}.$$

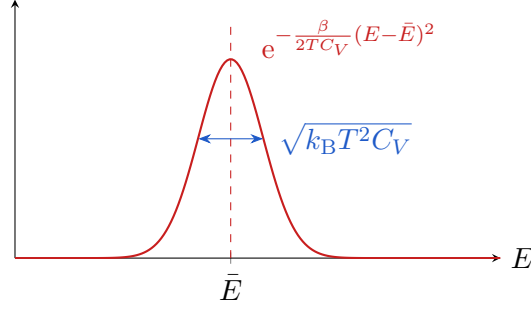


Figure 8.2: The energy distribution in the canonical ensemble is sharply peaked at $\bar{E}$, with width $\sqrt{k_B T^2 C_V}$.

8.7 Classical Ideal Gas

With

$$H = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m},$$

the partition function is

$$Q_N(T, V) = \int \frac{d^{3N}p d^{3N}q}{N! h^{3N}} e^{-\beta(\vec{p}_1^2 + \dots + \vec{p}_N^2)/2m} = \frac{V^N}{N!} \left(\int \frac{dp}{h} e^{-\beta p^2/2m} \right)^{3N} = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N,$$

where

$$\lambda := \sqrt{\frac{2\pi\hbar^2}{mk_B T}} \quad (8.2)$$

is the **thermal wavelength**. Then, using Stirling's approximation,

$$A(T, V) = -k_B T \ln Q_N = N k_B T \left[\ln(n\lambda^3) - 1 \right], \quad n = \frac{N}{V}.$$

9 Grand Canonical Ensemble

The system now exchanges both energy and particles with the reservoir. The relative probability of finding the system with N particles at temperature T in a volume V is taken to be

$$\rho(N, T, V) = z^N Q_N(T, V),$$

where

$$z := e^{\beta\mu}$$

is the **fugacity** and μ is the **chemical potential**.

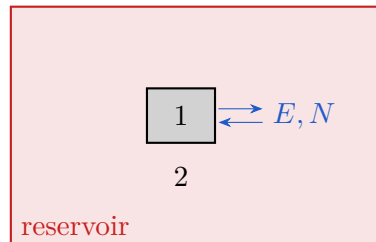


Figure 9.1: System 1 exchanges energy and particles with the reservoir 2.

9.2 Grand Partition Function

$$\mathcal{Q}(z, T, V) := \sum_{N=0}^{\infty} z^N Q_N(T, V).$$

The internal energy is

$$U = \frac{\sum_{N=0}^{\infty} E_N z^N Q_N}{\sum_{N=0}^{\infty} z^N Q_N}, \quad E_N = -\frac{\partial}{\partial \beta} \ln Q_N,$$

where E_N is the average energy of a canonical ensemble with N particles. At fixed z ,

$$U = -\frac{\partial}{\partial \beta} \ln \mathcal{Q}(z, T, V).$$

9.3 Number Fluctuations

The average number of particles is

$$\bar{N} = \frac{\sum_N N z^N Q_N}{\sum_N z^N Q_N} = z \frac{\partial}{\partial z} \ln \mathcal{Q}(z, T, V).$$

Differentiating once more,

$$z \frac{\partial}{\partial z} z \frac{\partial}{\partial z} \ln \mathcal{Q} = z \frac{\partial}{\partial z} \left[\frac{\sum_N N z^N Q_N}{\sum_N z^N Q_N} \right] = \overline{N^2} - \frac{(\sum_N N z^N Q_N)^2}{(\sum_N z^N Q_N)^2} = \overline{N^2} - \bar{N}^2.$$

Since $z = e^{\beta\mu}$, i.e. $\beta\mu = \ln z$,

$$z \frac{\partial}{\partial z} = z \frac{\partial \mu}{\partial z} \frac{\partial}{\partial \mu} = \frac{1}{\beta} \frac{\partial}{\partial \mu} \implies z \frac{\partial}{\partial z} z \frac{\partial}{\partial z} = \frac{1}{\beta^2} \frac{\partial^2}{\partial \mu^2},$$

so that

$$\overline{N^2} - \bar{N}^2 = (k_B T)^2 \frac{\partial^2}{\partial \mu^2} \ln \mathcal{Q}(z, T, V),$$

or, in terms of the density $n = N/V$,

$$\overline{n^2} - \bar{n}^2 = \left(\frac{k_B T}{V} \right)^2 \frac{\partial^2}{\partial \mu^2} \ln \mathcal{Q}(z, T, V).$$

In the thermodynamic limit $V \rightarrow \infty$ we expect $\frac{1}{V} \ln \mathcal{Q}(z, T, V)$ to tend to a finite limit, so that $\overline{n^2} - \bar{n}^2 \sim 1/V$ vanishes. Hence the grand canonical ensemble is equivalent to the canonical ensemble with $N = \bar{N}$.

9.4 Connection with Thermodynamics

Since the number fluctuations are vanishingly small, we need to keep only the largest term in the sum over N :

$$\begin{aligned} \ln \mathcal{Q}(z, T, V) &= \ln \left(\sum_N z^N Q_N(T, V) \right) \simeq \ln(z^{\bar{N}} Q_{\bar{N}}(T, V)) \\ &= \bar{N} \ln z + \ln Q_{\bar{N}}(T, V) = \frac{\bar{N} \mu}{k_B T} + \ln Q_{\bar{N}}(T, V). \end{aligned}$$

With $Q_{\bar{N}} = e^{-\beta A_{\bar{N}}}$ and, by extensivity, $A_N(V, T) = N a(v, T)$ with $v = V/N$,

$$\ln \mathcal{Q}(z, T, V) = \frac{\bar{N}}{k_B T} [\mu - a(v, T)].$$

From $dU = T dS - P dV + \mu dN$ we have $dA = -S dT - P dV + \mu dN$, so

$$P = -\frac{\partial A}{\partial V} = -N \frac{\partial a}{\partial V} = -\left(\frac{\partial a}{\partial v}\right)_T,$$

$$\mu = \left(\frac{\partial A}{\partial N}\right)_{V, T} = a + N \frac{\partial a}{\partial N} = a + N \frac{\partial v}{\partial N} \frac{\partial a}{\partial v} = a - v \frac{\partial a}{\partial v} = a + vP.$$

Therefore $\mu - a = vP$ and

$$\ln \mathcal{Q}(z, T, V) = \frac{PV}{k_B T}.$$

From now on the bar on $\bar{N}$ is omitted, $\bar{N} \equiv N$.

14 Quantum Statistics

14.1 Thermal Wavelength

Atoms are wave packets. As T goes down, the wave packets start to overlap and quantum effects must be taken into account. The spatial extension of a wave packet is given by the **de Broglie wavelength** $\lambda_0 = h/p_0$, with

$$\frac{p_0^2}{2m} = \frac{3}{2} k_B T \quad \Rightarrow \quad \lambda_0 = \frac{h}{\sqrt{3mk_B T}}.$$

Up to a numerical factor this is the **thermal wavelength** (8.2),

$$\lambda := \sqrt{\frac{2\pi\hbar^2}{mk_B T}}.$$

With the density $n = N/V$:

$$\begin{array}{ll} n\lambda^3 \ll 1 & \text{classical regime,} \\ n\lambda^3 \simeq 1 & \text{onset of quantum effects.} \end{array}$$

The **degeneracy temperature** T_0 is defined by $n\lambda^3(T_0) = 1$:

$$k_B T_0 = \frac{2\pi\hbar^2}{m} n^{2/3}.$$

14.3 Occupation Numbers

Let us consider a complete orthonormal set of single-particle wave functions $u_\alpha(\vec{r})$:

$$\int d^3r u_\alpha^*(\vec{r}) u_\beta(\vec{r}) = \delta_{\alpha\beta}, \quad \sum_\alpha u_\alpha(\vec{r}) u_\alpha^*(\vec{r}') = \delta(\vec{r} - \vec{r}').$$

The N -particle wave function is

$$\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) = \frac{1}{\sqrt{N!}} \sum_P \delta_P P[u_{\alpha_1}(\vec{r}_1) \cdots u_{\alpha_N}(\vec{r}_N)],$$

where the sum runs over all permutations P of the coordinates and

$$\text{bosons: } \delta_P = 1, \quad \text{fermions: } \delta_P = \begin{cases} +1, & P \text{ even,} \\ -1, & P \text{ odd.} \end{cases}$$

The number of particles in state α is the **occupation number** n_α :

$$n_\alpha = \begin{cases} 0, 1, 2, \dots & \text{for bosons,} \\ 0, 1 & \text{for fermions,} \end{cases} \quad \sum_\alpha n_\alpha = N.$$

In the thermodynamic limit the last condition becomes $\int \frac{d^3r d^3p}{h^3} n_{\vec{p}} = N$.

14.5 Microcanonical Ensemble

The cell occupation numbers n_i satisfy the constraints

$$\sum_i n_i = N, \quad \sum_i n_i \varepsilon_i = E.$$

The division into cells is an imprecise coarse-graining process.

To find the number of ways in which $\{n_i\}$ can be realised, we assign particles to single-particle quantum states without paying attention to which particle goes into which cell, since the exchange of particles in different cells does not produce a new state. Thus we can calculate the occupation of each cell independently and multiply the results:

$$\Omega\{n_i\} = \prod_j \omega_j(n_j),$$

where $\omega_j(n_j)$ is the number of ways to put n_j particles into the j -th cell, which contains g_j single-particle states.

14.6 Fermi Statistics

For fermions each single-particle state holds at most one particle, so

$$\omega_j(n_j) = \frac{g_j!}{n_j! (g_j - n_j)!}, \quad \Omega\{n_i\} = \prod_j \frac{g_j!}{n_j! (g_j - n_j)!}.$$

Then

$$\ln \Omega\{n_i\} = \sum_j [\ln(g_j!) - \ln(n_j!) - \ln((g_j - n_j)!)],$$

and, assuming $n_j \gg 1$ and $g_j \gg 1$,

$$\ln \Omega\{n_i\} \simeq \sum_j [g_j \ln g_j - n_j \ln n_j - (g_j - n_j) \ln(g_j - n_j)].$$

The most probable distribution $\{\bar{n}_j\}$ is obtained by maximising this subject to the constraints:

$$\delta \left[\ln \Omega\{n_i\} + \alpha \sum_j n_j - \beta \sum_j n_j \varepsilon_j \right] = 0,$$

which gives the **Fermi distribution**

$$\bar{n}_j = \frac{g_j}{e^{-\alpha + \beta \varepsilon_j} + 1}.$$

14.7 Bose Statistics

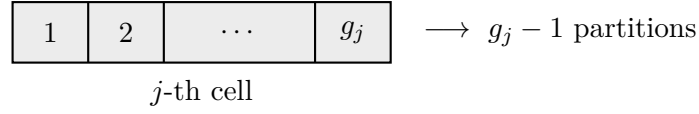


Figure 14.1: A cell with g_j single-particle states is separated by $g_j - 1$ partitions.

The number of ways to populate the cell with n_j bosons is found by throwing the n_j particles into the cell in any manner whatsoever, and then counting the number of distinct configurations obtainable when we permute the n_j particles together with the $g_j - 1$ partitions:

$$\omega_j(n_j) = \frac{(n_j + g_j - 1)!}{n_j! (g_j - 1)!}, \quad \Omega\{n_i\} = \prod_j \frac{(n_j + g_j - 1)!}{n_j! (g_j - 1)!}.$$

Maximising as before gives the **Bose distribution**

$$\bar{n}_j = \frac{g_j}{e^{-\alpha + \beta \varepsilon_j} - 1}.$$

14.8 Determining the Parameters

Both cases can be written together as

$$\bar{n}_j = \frac{g_j}{e^{-\alpha + \beta \varepsilon_j} \pm 1}, \quad \begin{cases} + & \text{Fermi,} \\ - & \text{Bose.} \end{cases}$$

Then

$$N = \sum_j \bar{n}_j = \sum_j \frac{g_j}{e^{-\alpha + \beta \varepsilon_j} \pm 1}.$$

Replacing the sum over cells by a sum over single-particle states $\vec{k}$, $\sum_j g_j(\dots) \rightarrow \sum_{\vec{k}}(\dots)$,

$$N = \sum_{\vec{k}} \frac{1}{e^{-\alpha + \beta \varepsilon_k} \pm 1}.$$

In the thermodynamic limit, with $\varepsilon_k = \hbar^2 k^2 / 2m$,

$$n = \int \frac{d^3k}{(2\pi)^3} \frac{1}{e^{-\alpha + \beta \varepsilon_k} \pm 1} = \frac{1}{2\pi^2} \int_0^\infty dk \frac{k^2}{e^{-\alpha + \beta \varepsilon_k} \pm 1}.$$

With $\beta \varepsilon_k = x^2$,

$$n = \frac{1}{2\pi^2} \left(\frac{2m}{\beta \hbar^2} \right)^{3/2} \int_0^\infty dx \frac{x^2}{e^{-\alpha + x^2} \pm 1}.$$

For n to stay finite as $\beta \rightarrow 0$ we need $e^{-\alpha} \rightarrow \infty$, so the ± 1 can be neglected and

$$\bar{n}_k \rightarrow e^{\alpha - \beta \varepsilon_k}.$$

To coincide with the Maxwell–Boltzmann distribution in this limit we need

$$\beta = \frac{1}{k_B T}.$$

The remaining parameter defines the **fugacity** z and the **chemical potential** μ ,

$$e^\alpha = z = e^{\beta \mu},$$

and the **occupation number of state** $\vec{k}$ is

$$\bar{n}_k = \frac{1}{z^{-1} e^{\beta \varepsilon_k} \pm 1}, \quad \begin{cases} + & \text{Fermi,} \\ - & \text{Bose.} \end{cases} \quad (14.1)$$

Grand canonical derivation. The same result follows from the grand partition function

$$\mathcal{Q} = \sum_{N=0}^{\infty} z^N Q_N, \quad z = e^{\beta\mu}.$$

For an ideal quantum gas the N -particle states are labelled by the occupation numbers, with energy $\sum_i n_i \varepsilon_i$ and $N = \sum_i n_i$.

Bosons.

$$\begin{aligned} \mathcal{Q} &= \sum_N z^N \sum_{\substack{\{n_i\} \\ \sum_i n_i = N}} e^{-\beta \sum_i n_i \varepsilon_i} = \sum_{\{n_i\}} \prod_i [ze^{-\beta \varepsilon_i}]^{n_i} \\ &= \sum_{n_0=0}^{\infty} (ze^{-\beta \varepsilon_0})^{n_0} \sum_{n_1=0}^{\infty} (ze^{-\beta \varepsilon_1})^{n_1} \dots = \frac{1}{1 - ze^{-\beta \varepsilon_0}} \frac{1}{1 - ze^{-\beta \varepsilon_1}} \dots, \end{aligned}$$

so the **bosonic grand partition function** is

$$\mathcal{Q}(z, V, T) = \prod_{\vec{k}} (1 - ze^{-\beta \varepsilon_k})^{-1}.$$

Fermions. Now $n_i \in \{0, 1\}$:

$$\mathcal{Q}(z, V, T) = \sum_{n_0=0}^1 (ze^{-\beta \varepsilon_0})^{n_0} \sum_{n_1=0}^1 (ze^{-\beta \varepsilon_1})^{n_1} \dots = (1 + ze^{-\beta \varepsilon_0})(1 + ze^{-\beta \varepsilon_1}) \dots,$$

so the **fermionic grand partition function** is

$$\mathcal{Q}(z, V, T) = \prod_{\vec{k}} (1 + ze^{-\beta \varepsilon_k}).$$

Both can be written as

$$\mathcal{Q}(z, V, T) = \prod_{\vec{k}} (1 \mp ze^{-\beta \varepsilon_k})^{\mp 1}, \quad \begin{cases} - & \text{bosons,} \\ + & \text{fermions.} \end{cases}$$

The average occupation of state $\vec{k}$ is

$$\begin{aligned} \langle n_k \rangle &= \frac{1}{\mathcal{Q}} \sum_{\{n_i\}} n_k z^{\sum_i n_i} e^{-\beta \sum_i n_i \varepsilon_i} = -\frac{1}{\beta} \frac{\partial}{\partial \varepsilon_k} \ln \mathcal{Q}(z, V, T) \\ &= \pm \frac{1}{\beta} \frac{\partial}{\partial \varepsilon_k} \sum_{\vec{j}} \ln(1 \mp ze^{-\beta \varepsilon_j}) = \frac{ze^{-\beta \varepsilon_k}}{1 \mp ze^{-\beta \varepsilon_k}}, \end{aligned}$$

that is,

$$\langle n_k \rangle = \frac{1}{z^{-1} e^{\beta \varepsilon_k} \mp 1}, \quad \begin{cases} - & \text{bosons,} \\ + & \text{fermions,} \end{cases}$$

in agreement with (14.1).

Finally, the total number of particles is

$$N = \sum_{\vec{k}} \langle n_k \rangle \simeq \frac{V}{(2\pi)^3} \int d^3k \langle n_k \rangle = \frac{V}{(2\pi)^3} \int dE \int d^3k \frac{\delta(\varepsilon(\vec{k}) - E)}{z^{-1} e^{\beta E} \mp 1},$$

with $d^3k = 4\pi k^2 dk$ for an isotropic dispersion.